

AN INTERNSHIP REPORT
OF
Data Science Master Virtual Internship

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement for the degree of
B.Tech (Computer Science and Engineering)

Submitted By

Roll Number: 2410030714
Name of Student: Tejas Aditya
Batch: 2024 - 2028

1. Candidate's Declaration

I, Tejas Aditya, Roll Number 2410030714, hereby declare that this Internship Report titled "Data Science Master Virtual Internship" is my own original work, carried out by me during the period 10 Weeks (January 2026 - March 2026), and has not been submitted, in part or in full, to any other university or institution for the award of any degree or diploma.

I further declare that the information and content presented in this report are based on my genuine learning experience, official communication, and certificates received during the course of the internship, and have been compiled in fulfilment of the requirements for the degree of B.Tech (Computer Science and Engineering) at IILM University, Greater Noida, U.P..

Name of Student: Tejas Aditya

Roll Number: 2410030714

Batch: 2024 - 2028

Place: Greater Noida

Date: _____

2. Acknowledgement

I would like to express my sincere gratitude to the AICTE-EduSkills National Internship Portal, a Ministry of Education initiative, and to Altair, the technology partner supporting the "Data Science Master Virtual Internship", for providing me with the opportunity to undertake this internship and for granting access to a well-structured, industry-oriented curriculum built on the RapidMiner platform.

I am thankful to the mentors and facilitators associated with the EduSkills program, and to Mr. Bibek Ranjan, Director, EduSkills, for coordinating this valuable virtual internship initiative and for ensuring that the learning modules, assessments, and certification process were conducted smoothly.

I extend my heartfelt thanks to IILM University, Greater Noida, U.P. and the School of Computer Science and Engineering for encouraging students to pursue industry-recognized internships as part of the academic curriculum, and for providing the guidance necessary to document this experience formally.

I am also grateful to the faculty members of the School of Computer Science and Engineering for their constant support, valuable suggestions, and encouragement throughout the internship and the preparation of this report.

Finally, I wish to thank my parents and friends for their unwavering support and motivation, which enabled me to complete this internship successfully and prepare this report to the best of my ability.

Tejas Aditya

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4. Internship Completion Certificate

This section presents the official documentation confirming enrolment in and completion of the internship program. The following details have been extracted from the internship offer letter and the verified completion certificate:

Particular	Detail
Internship Title	Data Science Master Virtual Internship
Internship Domain	Data Science (Master Level)
Offer Letter Reference Number	C15Y2026M081350404
AICTE Student ID	STU69930cc624e851771244742
Certificate ID	410db09a43090949ea99
Duration	10 Weeks (January 2026 - March 2026)
Program	AICTE-EduSkills Virtual Internship, supported by Altair
Grade	O - Outstanding



Ref No: C15Y2026M081350404

Internship Offer Letter

Student Name : Tejas Aditya
Stream : B.Tech
Branch : B.Tech(Computer Science and Engineering (Cyber Security))
Institute Name : IILM University, Greater Noida
Internship Domain : Siemens Data Science Master
Internship Duration : January 2026 - March 2026
AICTE Student ID : STU69930cc624e851771244742

We are pleased to inform you that you have been selected for the 10-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.



शिक्षा मंत्रालय
MINISTRY OF
EDUCATION



अखिल भारतीय तकनीकी शिक्षा परिषद्
All India Council for Technical Education



EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

Tejas Aditya

IILM University, Greater Noida

has successfully completed the 10-weeks

Data Science Master Virtual Internship

During January - March 2026

May this Internship learning propel you toward a bright and successful career.

Supported by  **ALTAIR**

Ramesha B.S.
Head - Academic Initiatives
Altair

Dr. Buddha Chandrasekhar
Chief Coordinating Officer (CCO)
AICTE, Ministry of Education

Shubhajit Jagadev
Chief Executive Officer (CEO)
EduSkills



Certificate ID : 410db09a43090949ea99

Student ID: STU69930cc624e851771244742



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

The following module-wise certifications were earned upon successful completion of the respective learning tracks and are attached as supporting proof of completion:

- Applications & Use Cases Professional Certification



- Applications & Use Cases Master Certification



- Data Engineering Professional Certification



This is to certify that

Tejas Aditya

successfully passed the examination for

Data Engineering Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Data Access, Basic Transformations, Working with Multiple Data Sets, Pivot Tables, Routines and Simple Text Processing.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

- Data Engineering Master Certification



This is to certify that

Tejas Aditya

successfully passed the examination for

Data Engineering Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Loops and Branches, Advanced Text Processing, Exception Handling, Logging, Data Cleansing, Regular Expressions, Macros, Web APIs, and Scripting.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

- Machine Learning Professional Certification



This is to certify that

Tejas Aditya

successfully passed the examination for

Machine Learning Professional Certification

The Professional level is the first of two certification levels and is rewarded for solid experience in the topic. Some of the topics tested at this level are:

Classification, Regression, Scoring, Hold-out Validation, Correlations, Feature Importance, Clustering and Association Rules.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

- Machine Learning Master Certification



This is to certify that

Tejas Aditya

successfully passed the examination for

Machine Learning Master Certification

The Master level is the second level after Professional and stands for high proficiency in the topic. Some of the topics tested at this level are:

Complex Predictive Models, Cross Validation and Correct Validation, Parameter Optimization, Model Selection, Feature Engineering, Time Series and Forecasting and Integrating R and Python Models.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

- Platform Administration Master Certification



This is to certify that

Tejas Aditya

successfully passed the examination for

Platform Administration Master Certification

The Master level stands for high proficiency in the topic. Some of the topics tested at this level are:

Platform Overview, RapidMiner Studio Installation, RapidMiner AI Hub Administration, RapidMiner Radoop, RapidMiner Real-Time Scoring, and RapidMiner Marketplace.

This certificate can be verified using the corresponding open badge which was issued by Altair Engineering Inc. to the certificate holder. The certificate holder can provide a link to the validated badge on openbadgepassport.com or another open badge service.


Knut Makowski
(Head of Education)

5. Project Description

5.1 Introduction

This report documents my internship experience under the "Data Science Master Virtual Internship", undertaken as part of the B.Tech CSE curriculum to build practical, industry-relevant skills in data science. The internship was delivered over a structured 10 Weeks (January 2026 - March 2026) virtual program through the AICTE-EduSkills National Internship Portal, a Ministry of Education initiative, and was supported by Altair, the developer of the RapidMiner data science platform on which the entire curriculum was built.

The program progressed through data engineering, machine learning, and applied use-cases, culminating in a verified internship credential graded "O - Outstanding". Learning was organized around the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology, giving the curriculum a coherent, end-to-end structure rather than a set of disconnected topics.

Through this internship, I gained practical exposure to accessing and cleansing raw data, engineering features, building and validating classification, regression, and clustering models, optimizing model parameters, and deploying models for real-time scoring. I also developed familiarity with platform administration on RapidMiner AI Hub and Radoop, and with integrating R and Python into RapidMiner workflows.

The primary learning objective of the internship was to bridge the gap between theoretical statistics taught in the classroom and the integrated, end-to-end data science workflow expected in industry roles - spanning data access, cleansing, modelling, validation, and deployment - using RapidMiner as the practical vehicle for skill acquisition.

5.2 Organization Profile

The AICTE-EduSkills National Internship Portal is an official placement platform operated under the All India Council for Technical Education (AICTE), a statutory body of the Ministry of Education, Government of India. The portal is designed to connect engineering students with structured, industry-aligned virtual internship programs delivered in partnership with recognized technology companies.

This internship was delivered in partnership with Altair, the company behind the RapidMiner data science and machine learning platform, which is used widely across enterprises for data preparation, predictive analytics, and model deployment. EduSkills, functioning as the program facilitator, was responsible for coordinating enrolment, structuring the week-wise curriculum, administering assessments, and issuing the verified completion certificate.

Through this collaborative model involving the Ministry of Education, AICTE, EduSkills, and Altair, the program contributes to bridging the academia-industry skill gap by providing students with structured access to an enterprise-grade platform, curated learning content, and a

certification pathway that is recognized as evidence of applied technical competence in the data science domain.

5.3 Problem Statement

Organizations across every sector now generate vast volumes of data, driving strong demand for professionals who can turn that data into reliable, actionable insight - from cleaning and structuring raw data to building and deploying predictive models. Academic curricula, however, often introduce data science concepts in isolated modules, leaving a gap between theoretical statistics and the integrated, end-to-end workflow - data access, cleansing, modelling, validation, and deployment - expected in industry roles.

This internship was undertaken to bridge that gap by building a single, connected understanding of the full data science lifecycle, from raw data to a validated, deployable model, using a structured curriculum spanning data engineering, statistical modelling, machine learning, and platform administration on the RapidMiner platform.

5.4 Project Objectives

- To build strong foundations in data access, transformation, and data cleansing techniques.
- To apply advanced data engineering techniques, including scripting, regular expressions (regex), macros, and Web API integration.
- To master core machine learning techniques such as classification, regression, clustering, and scoring.
- To develop complex predictive models employing cross-validation and parameter optimization.
- To apply the CRISP-DM methodology to real-world use cases and data visualization.
- To gain platform administration skills across the RapidMiner (Altair) ecosystem, including RapidMiner AI Hub and Radoop.
- To understand real-time model scoring and deployment strategies for production-like environments.
- To develop working familiarity with integrating R and Python models within RapidMiner workflows.

5.5 Scope of the Project

The scope of this internship was limited to simulation-based and platform-guided training conducted entirely in a virtual environment, using RapidMiner Studio and RapidMiner AI Hub as provided through the AICTE-EduSkills program infrastructure. The internship covered the full spectrum of data science operations - from professional-level and master-level modules in data engineering and machine learning, to applied use-cases and platform administration - all structured around the CRISP-DM methodology.

Practically, the internship involved working through guided exercises and scenario-based problems within each module, reinforced through hold-out and cross-validation exercises, and concluding with a final assessment. It did not involve deployment on a live organizational production system or client-facing project delivery; rather, it followed a structured virtual training and certification model designed to develop and validate individual competence in data science and machine learning tools.

The boundaries of the internship were defined by the 10 Weeks (January 2026 - March 2026) curriculum prescribed by AICTE-EduSkills and Altair, and all technical work was confined to the RapidMiner platform environment as provided through the virtual internship portal, concluding with a verified credential graded "O - Outstanding".

5.6 Technologies and Tools Used

- RapidMiner Studio - for data access, transformation, and model building
- RapidMiner AI Hub - for model deployment, real-time scoring, and platform administration
- Radoop - for big data processing within the RapidMiner ecosystem
- CRISP-DM Methodology - for structuring end-to-end data science workflows
- Data Cleansing, Transformation and Text Processing
- Scripting, Regex, Macros and Web API Integration
- Classification, Regression and Clustering techniques
- Feature Engineering and Cross-Validation
- Time Series Forecasting
- R and Python Integration
- Data Visualization

5.7 System Architecture

The internship's learning framework followed the CRISP-DM pipeline, representing how data moves from raw source to a deployed, scored model within the RapidMiner ecosystem:

Data Access Layer (Sourcing & ETL)



Cleansing & Transformation Layer



Feature Engineering Layer (Selection & Preparation)



Model Building Layer (Classification, Regression, Clustering)



Validation & Optimization Layer (Cross-Validation)



Deployment & Scoring Layer (Real-Time Scoring, RapidMiner AI Hub)

Data Access Layer: Involves sourcing and loading raw datasets through ETL (Extract, Transform, Load) operations, forming the starting point of every workflow built during the internship.

Cleansing & Transformation Layer: Covers cleansing, transforming, and structuring multi-table data, along with text processing, regex, and macros, to make datasets suitable for modelling.

Feature Engineering Layer: Focuses on selecting, engineering, and preparing features to improve model quality and predictive performance.

Model Building Layer: Encompasses classification, regression, clustering, and time-series forecasting to derive predictive insights from prepared data.

Validation & Optimization Layer: Applies hold-out and cross-validation techniques along with parameter optimization to ensure robust, generalizable model performance.

Deployment & Scoring Layer: Focuses on real-time scoring, monitoring, and management of trained models through RapidMiner AI Hub and Radoop, applying the CRISP-DM framework end to end.

5.8 Methodology

The internship followed a structured, phase-wise methodology spanning 10 Weeks (January 2026 - March 2026), progressing from foundational data engineering concepts to advanced machine learning and platform administration, concluding with a final assessment and verified credential.

Week	Module / Topic	Description
Week 1-2	Data Engineering - Professional	Data access, basic transformations, multiple data sets, pivot tables and simple text processing.
Week 3-4	Data Engineering - Master	Loops, branches, advanced text processing, exception handling, regex, macros and Web APIs.
Week 5-6	Machine Learning - Professional	Classification, regression, scoring, hold-out validation, correlations and clustering.
Week 7	Machine Learning - Master	Complex predictive models, cross-validation, parameter optimization and time series forecasting.
Week 8	Applications & Use Cases	CRISP-DM methodology, use cases for machine learning, visualization and model deployment strategy.
Week 9-10	Platform Administration & Wrap-Up	RapidMiner AI Hub, Radoop, real-time scoring, marketplace and final assessment.

5.9 Expected Outcomes

- Gained an integrated, end-to-end understanding of the data science lifecycle, from raw data to a deployed model.
- Developed confidence in data engineering: accessing, cleansing, and transforming data using RapidMiner.
- Learned to build and validate predictive models using classification, regression, and clustering techniques.
- Understood parameter optimization, feature engineering, and cross-validation for robust model performance.
- Practiced applying the CRISP-DM methodology to structure real-world data science use cases.
- Built familiarity with platform administration - RapidMiner AI Hub, Radoop, and real-time model scoring.
- Acquired hands-on exposure to time series forecasting and integration of R and Python models within RapidMiner workflows.
- Strengthened problem-solving ability and industry-tool proficiency, validated through a verified credential graded "O - Outstanding".

5.10 Certificates of Completion and Communication Proof

Copies of the official internship offer letter (Reference Number C15Y2026M081350404), the verified internship completion certificate (Certificate ID 410db09a43090949ea99, Grade "O - Outstanding"), and the module-wise certifications earned (Applications & Use Cases, Data Engineering, Machine Learning, and Platform Administration) are attached in Section 4 of this report as evidence of successful participation and completion of the internship program.

6. Bibliography / References

1. Altair Engineering Inc. - Official Website, <https://www.altair.com>
2. AICTE - All India Council for Technical Education, <https://www.aicte-india.org>
3. EduSkills Foundation - Virtual Internship Program, <https://eduskillsfoundation.org>
4. RapidMiner Documentation, <https://docs.rapidminer.com>
5. RapidMiner AI Hub Documentation, <https://docs.rapidminer.com/latest/aihub/>
6. IILM University, Greater Noida - Official Website, <https://iilm.edu>